

Weekly Update

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1 Main Findings

1.1 DB-Sorter

quick background. DB-sorter uses the sorting characteristic of DB flow to find ground state or excited states of a hamiltonian H using a diagonal matrix D that has an engineered order in its diagonal elements (usually increasing). We compare three different diagonals in terms of spectral gaps and synthesis complexity.

Findings:

- I simulated the method for three other hamiltonian models: TFIM, XXZ, and J1/J2 model. I tried three different diagonal choices for D . Cool down happens in all. See [GitHub](#) for code and [3.1.1](#)
- I investigated the scaling of the method and didn't seem promising. See [GitHub](#) for code and [3.1.2](#)
- I checked to see if DB-sorter is a reasonable method to warm-start VQE. Results are inconclusive at the moment. See [GitHub](#) for code
- While in theory DB-sorter is supposed to prepare the ground/excited states, it acts well as a cool-down method in general.
- In all the above tests I noticed that the base in which D is diagonal is very important. Intuitively, if D is diagonal in the same eigenbasis of H , we have the sharpest energy drop.

2 Plan For Next Week(s)

2.1 DB-sorter

- Wrapping up DB-sorter.

3 Details

3.1 DB-sorter

3.1.1 Testing DB-sorter with different D 's on different hamiltonian models

Previously, we saw that the energy gap of the diagonal, sets an upper bound on the convergence speed. Accordingly, we worked with three different diagonals:

- $D = -|0\rangle\langle 0|$: The projection has the biggest spectral gap, while highest degeneracy and deepest circuit ($\mathcal{O}(n)$)
- Monotone D : as defined originally in the first introduction of DB-sorter has zero degeneracy and $\mathcal{O}(1)$ depth and the worst spectral gap of $\mathcal{O}(2^{-n})$
- Select-K as defined in the previous update: Something in between: while maintaining $\mathcal{O}(1)$ depth, has lower degeneracy than the projection as a better spectral gap than the monotone choice; $\mathcal{O}(n^{-1})$

I tested our method with all of the choices above for D on TFIM, XXZ, and J1/J2 models. You can see the results in [Figure 1](#)

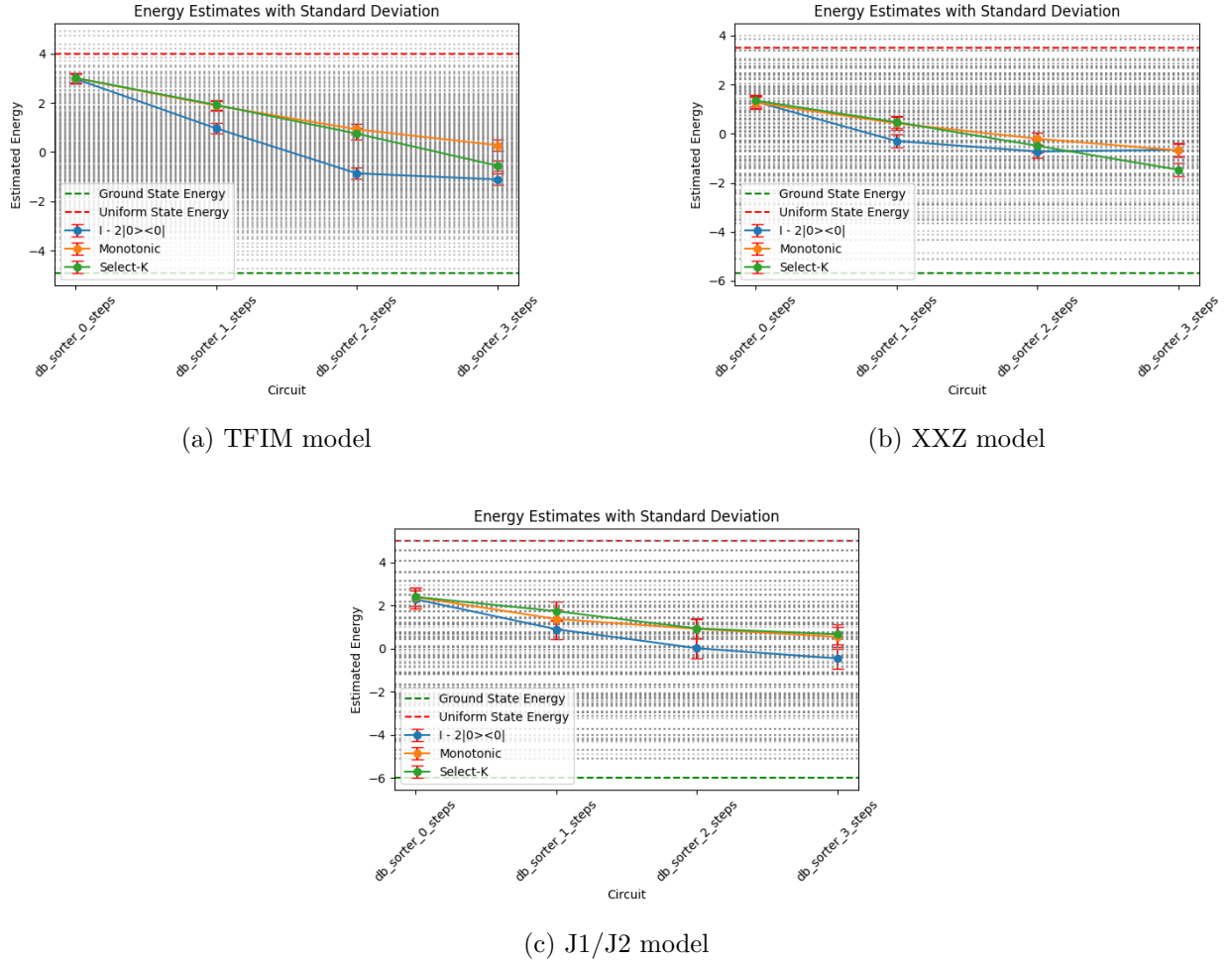
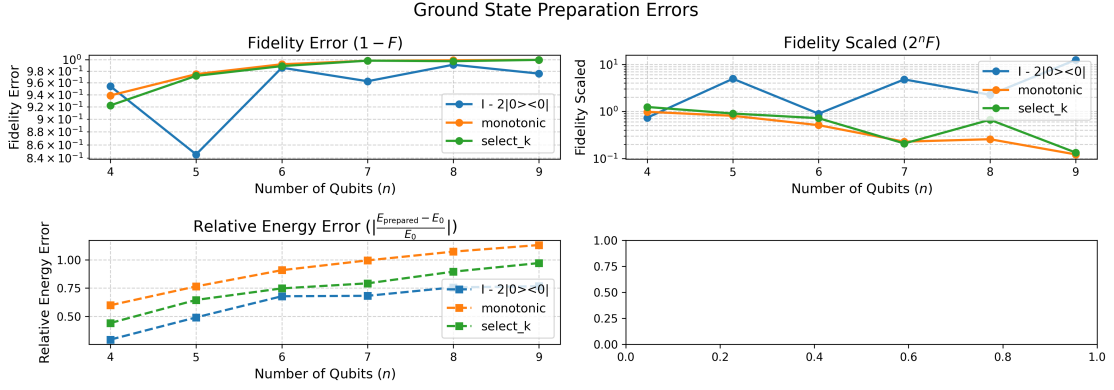


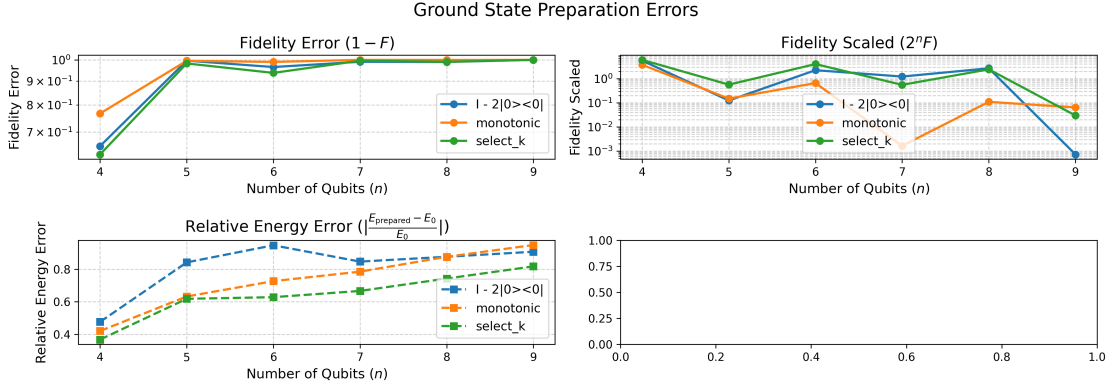
Figure 1: All models show a cool down behaviour. Projection consistently starts sharper and faster and for most cases select-k, as expected, ends up second. We had similar pattern for H_2 and NH_3

3.1.2 Scaling with the number of qubits

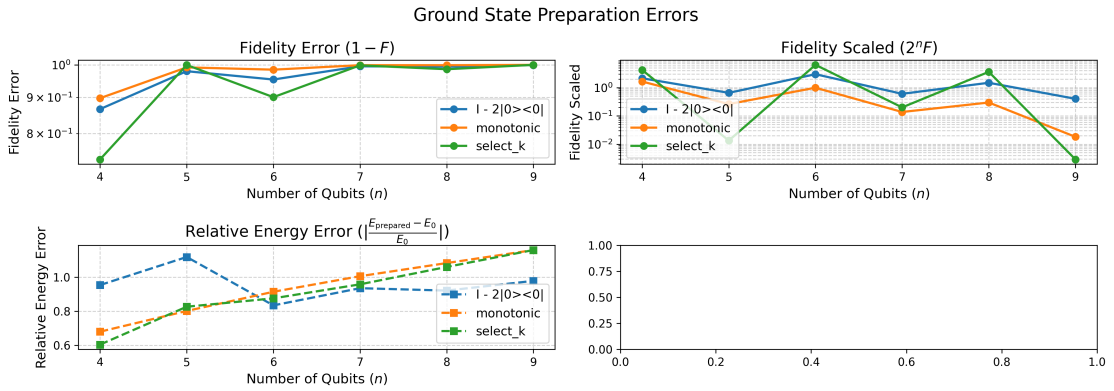
Scaling (and fidelity in general) did not prove to be promising with cold-start. You can see the results in [Figure 2](#)



(a) TFIM model



(b) XXZ model



(c) J1/J2 model

Figure 2: The fidelity is not promising with cold-start